

Softmax Offset

production, 5-member ensemble, evaluated on 2016-2026, measured against the market

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Model identity

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Softmax Offset production	EVALUATION YEARS 11 2016-2026	ENSEMBLE MEMBERS 5 seeds 7, 17, 42, 123, 256	BOOSTING ROUNDS 400 3-fold chronological CV	FEATURE SET production 28 declared inputs
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HOW EACH EVALUATION YEAR WAS FITTED

The 2016 evaluation model was trained on the initial 2010-2015 set. Each later evaluation adds the preceding complete season, so the training window expands by one year; the 2026 model uses 2010-2025. Every evaluation year is fitted separately -- 11 fits in all -- and no evaluation year's outcomes appear in its own fit.

Deltas in this report use `model_minus_market_log_loss: race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Provenance -- the source commit, the config and data digests, the environment and every input this report read by content digest -- is in the technical appendix and in `run_lock.json` beside this file. Canonical publication fails without it.

Executive summary

MEAN DELTA VS MARKET -0.004533 negative is better	RACES 7,933	YEARS BETTER 10/11	RACE-LEVEL WIN RATE 54.3%
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Softmax Offset is better than the market by 0.004533 `race_multinomial_log_loss` on 7,933 races. The direction holds in 10 of 11 evaluated years. Race by race it is ahead 54.3% of the time, which is the more honest figure: a mean delta can be carried by a few races. 3,638 bets were selected under the staking rules below, on retrospective odds that could not have been bet into.

Evaluation population and market reference

THE MARKET REFERENCE

Every metric below is reported beside the market's own implied probabilities on the same races, normalised within each race, and the delta is this model's race log loss minus the market's on the same races. One yardstick across every published report, so their headline numbers mean the same thing and can be read against one another. The market is a measurement, not a model this one is defined against.

THE EVALUATION POPULATION

Every figure in this report covers the standardized evaluation window, which begins in 2016. One window for every published report, so their headline numbers are computed on the identical race keys and can be compared -- which is checked by digest before anything is published, not asserted in a footnote. 2015 is earlier than the window and is excluded from every evaluation figure below, including the bankroll paths. The feature-effect grid is the one exception, and says so where it appears: a learned response direction is a property of each year's fit rather than of the evaluation population.

TEST RACES 7,933	YEARS 11 2016-2026
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Splits present: `test`, `train_apparent`, `train_prequential`. Only the forward test split is scored above. The others exist to be compared against it -- an apparent-fit number looks far better than a forward one -- and are never pooled with it.

Out-of-fold margin sources: `market_fallback`, `model`.

One or more years is partial

2026 does not span a full season, so its per-year figures rest on fewer races than the others. Partial years are marked in every table rather than dropped, because dropping them would silently change the evaluation population.

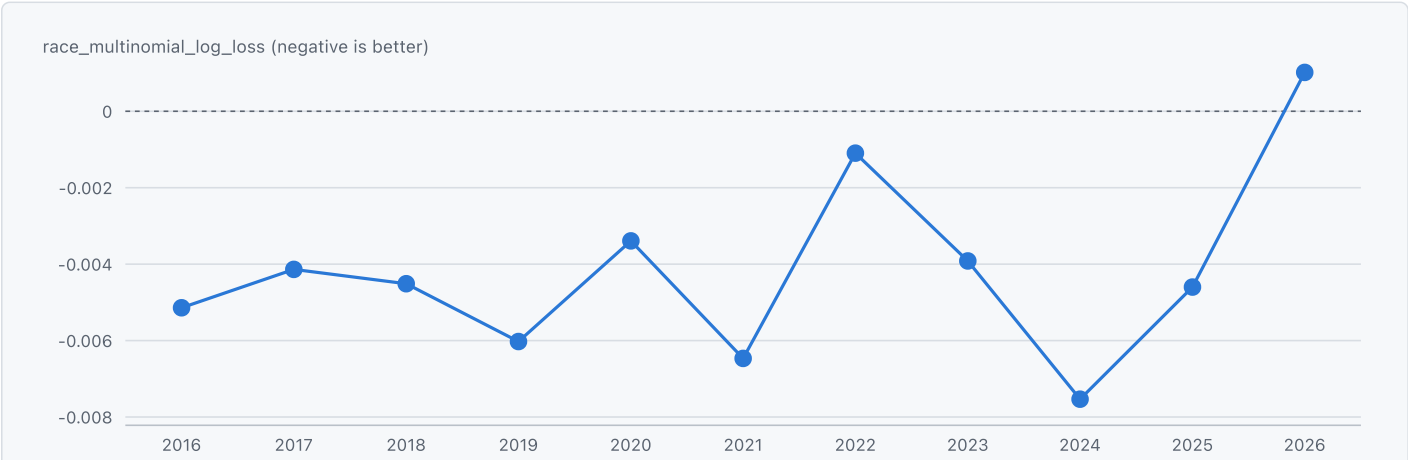
Overall probability performance

races	model log loss	market log loss	model minus market log loss	better race share
7933	2.024130	2.028662	-0.004533	0.542670

One row, pooled over every evaluated race, and measured against the market: `model_minus_market_log_loss` is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	model minus market log loss	better race share	is partial year
2016	744	2.094135	2.099277	-0.005142	0.564516	no
2017	772	2.017423	2.021559	-0.004136	0.528497	no
2018	756	2.056990	2.061504	-0.004514	0.525132	no
2019	759	1.926801	1.932827	-0.006026	0.548090	no
2020	799	2.007010	2.010400	-0.003390	0.535670	no
2021	783	2.000193	2.006659	-0.006466	0.565773	no
2022	783	2.050100	2.051195	-0.001095	0.519796	no
2023	796	2.020147	2.024066	-0.003919	0.530151	no
2024	792	2.018210	2.025744	-0.007534	0.565657	no
2025	748	2.052795	2.057397	-0.004601	0.554813	no
2026	201	2.027241	2.026224	0.001016	0.502488	yes



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2016	-0.005142
2017	-0.004136
2018	-0.004514
2019	-0.006026
2020	-0.00339
2021	-0.006466
2022	-0.001095
2023	-0.003919
2024	-0.007534
2025	-0.004601
2026	0.001016

Ahead of the market in 10 of 11 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

Reliability by pipeline stage, on bins fixed across the whole report so two stages are comparable, pooled from the per-year counts rather than averaged over years. Stages: market, raw_offset, temperature, market_blend, bagged.

MARKET

Pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	10,889	36	0.0062	0.0033	0.538
0.01-0.03	16,603	263	0.0171	0.0158	0.925
0.03-0.05	17,886	625	0.0365	0.0349	0.956
0.05-0.07	15,094	901	0.0625	0.0597	0.955
0.07-0.10	10,318	843	0.0874	0.0817	0.935
0.10-0.15	11,807	1,511	0.1223	0.1280	1.046
0.15-0.20	6,323	1,130	0.1728	0.1787	1.034
0.20-0.30	5,668	1,389	0.2419	0.2451	1.013
0.30-0.50	2,723	1,040	0.3651	0.3819	1.046
0.50-1.00	325	195	0.5566	0.6000	1.078

RAW_OFFSET

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	63,732	302	0.0057	0.0047	0.825
0.01-0.03	82,739	1,351	0.0171	0.0163	0.955
0.03-0.05	91,449	3,421	0.0368	0.0374	1.018
0.05-0.07	67,420	4,116	0.0620	0.0611	0.985
0.07-0.10	49,563	4,205	0.0866	0.0848	0.980
0.10-0.15	55,886	7,182	0.1224	0.1285	1.050
0.15-0.20	30,529	5,325	0.1729	0.1744	1.009
0.20-0.30	29,613	7,042	0.2422	0.2378	0.982
0.30-0.50	15,233	5,545	0.3687	0.3640	0.987
0.50-1.00	2,016	1,176	0.5688	0.5833	1.026

TEMPERATURE

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	66,653	337	0.0056	0.0051	0.897
0.01-0.03	82,703	1,406	0.0171	0.0170	0.994
0.03-0.05	90,570	3,432	0.0367	0.0379	1.032
0.05-0.07	66,792	4,091	0.0620	0.0612	0.989
0.07-0.10	48,766	4,179	0.0866	0.0857	0.990
0.10-0.15	55,018	7,072	0.1224	0.1285	1.050
0.15-0.20	30,123	5,226	0.1728	0.1735	1.004
0.20-0.30	29,618	7,033	0.2421	0.2375	0.981
0.30-0.50	15,695	5,608	0.3694	0.3573	0.967
0.50-1.00	2,242	1,281	0.5721	0.5714	0.999

MARKET_BLEND

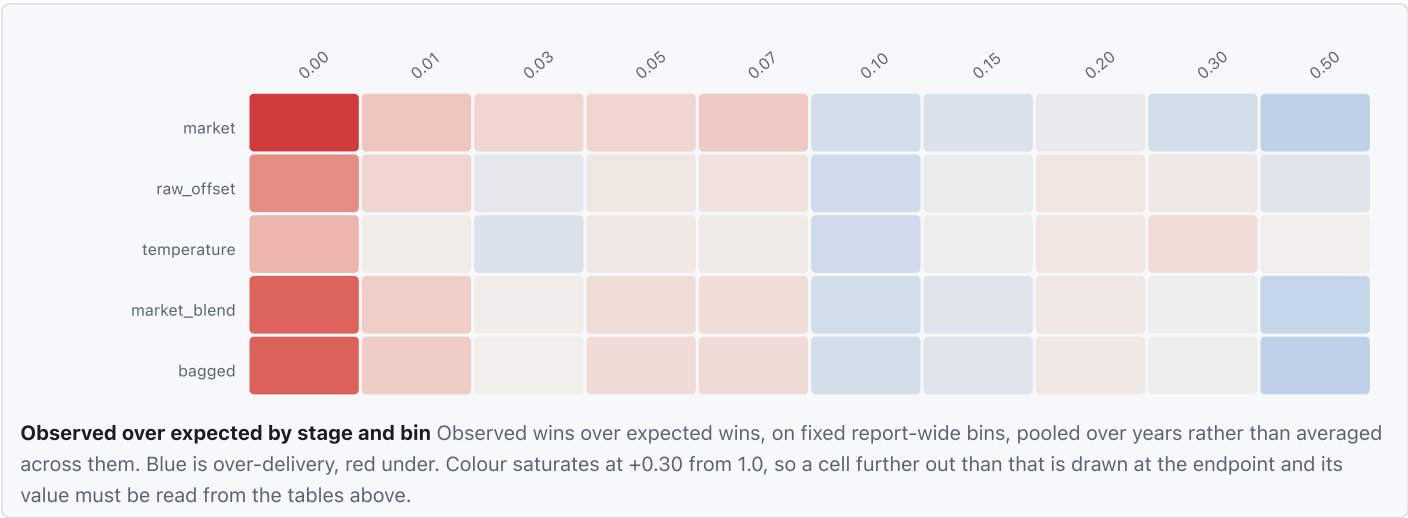
Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	61,214	271	0.0059	0.0044	0.756
0.01-0.03	82,858	1,333	0.0171	0.0161	0.941
0.03-0.05	91,049	3,336	0.0368	0.0366	0.996
0.05-0.07	67,103	4,015	0.0619	0.0598	0.967
0.07-0.10	52,718	4,401	0.0863	0.0835	0.968
0.10-0.15	56,434	7,232	0.1224	0.1281	1.047
0.15-0.20	30,803	5,456	0.1726	0.1771	1.026
0.20-0.30	29,419	7,039	0.2422	0.2393	0.988
0.30-0.50	14,685	5,434	0.3687	0.3700	1.004
0.50-1.00	1,897	1,148	0.5666	0.6052	1.068

BAGGED

The bagged probabilities, pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	12,231	54	0.0059	0.0044	0.754
0.01-0.03	16,601	266	0.0171	0.0160	0.937
0.03-0.05	18,193	670	0.0368	0.0368	1.001
0.05-0.07	13,423	801	0.0619	0.0597	0.964
0.07-0.10	10,529	877	0.0863	0.0833	0.965
0.10-0.15	11,278	1,443	0.1223	0.1279	1.046
0.15-0.20	6,179	1,096	0.1725	0.1774	1.028
0.20-0.30	5,884	1,407	0.2421	0.2391	0.988
0.30-0.50	2,940	1,088	0.3687	0.3701	1.004
0.50-1.00	378	231	0.5667	0.6111	1.078



The largest departure is the 0.000 to 0.010 bin, and every stage misses there: market 0.54, raw_offset 0.83, temperature 0.90, market_blend 0.76, bagged 0.75. market is furthest from 1.0 at 0.54, meaning that bin won fewer races than it priced; temperature is closest at 0.90, so this bin is where the pipeline moves calibration most. On the market stage that bin holds 10,889 runners, the 6th smallest of 10 bins, so the departure is not a thin-bin artefact. Calibration in this bin does not improve monotonically through the pipeline: temperature reaches 0.90 and market_blend falls back to 0.76, so that step gives back part of the correction rather than adding to it.

SHARPNESS

stage	races	mean entropy	mean normalised entropy	mean top probability	mean field size
market	7,933	2.0688	0.8274	0.2907	12.3076
raw_offset	7,933	2.0310	0.8124	0.3035	12.3076
temperature	7,933	2.0196	0.8079	0.3075	12.3076
market_blend	7,933	2.0416	0.8166	0.3000	12.3076
bagged	7,933	2.0417	0.8167	0.3000	12.3076

Entropy normalised by field size, so an eight-runner race and a fourteen-runner race are on one scale, and averaged over years weighted by races rather than as a mean of yearly means. A sharper distribution is not automatically better: it is only better if the calibration above holds.

Top-choice movement

test year	n races	n disagreement races	disagreement share	model top win rate	market top win rate	disagreement mean delta	agreement mean delta	seed top choice agreement
2016	744	52	0.069892	0.250000	0.250000	-0.013700	-0.004499	0.965385
2017	772	56	0.072539	0.321429	0.107143	-0.017866	-0.003062	0.871429
2018	756	60	0.079365	0.183333	0.050000	-0.001932	-0.004737	0.930000
2019	759	50	0.065876	0.300000	0.220000	-0.021397	-0.004942	0.928000
2020	799	43	0.053817	0.209302	0.186047	0.003582	-0.003787	0.902326
2021	783	70	0.089400	0.242857	0.257143	-0.017521	-0.005381	0.931429
2022	783	50	0.063857	0.240000	0.220000	-0.006338	-0.000737	0.892000
2023	796	45	0.056533	0.177778	0.222222	-0.008206	-0.003662	0.924444
2024	792	50	0.063131	0.160000	0.260000	0.004092	-0.008317	0.892000
2025	748	59	0.078877	0.203390	0.169492	-0.006463	-0.004442	0.911864
2026	201	11	0.054726	0.090909	0.363636	0.006035	0.000726	0.909091

How often this model's first pick differs from the market's, and whether it is right when it does. Disagreeing more is not better on its own -- the useful figure is the win rate *conditional* on disagreeing, because a model that disagrees often and loses is worse than one that never disagrees. The last column is how often the seeds pick the same runner as each other, which bounds how much of any disagreement is signal rather than seed noise.

Ensemble stability

RUNNERS COVERED

97,636

YEARS

11

SEEDS BAGGED

5

Member dispersion per runner. The bag is only worth its cost if members genuinely disagree: an ensemble whose members agree everywhere is one fit with five times the runtime. The per-runner spread across members is in `ensemble_diagnostics.parquet`, race by race.

Feature and correction effects

These are corrections, not drivers of winning

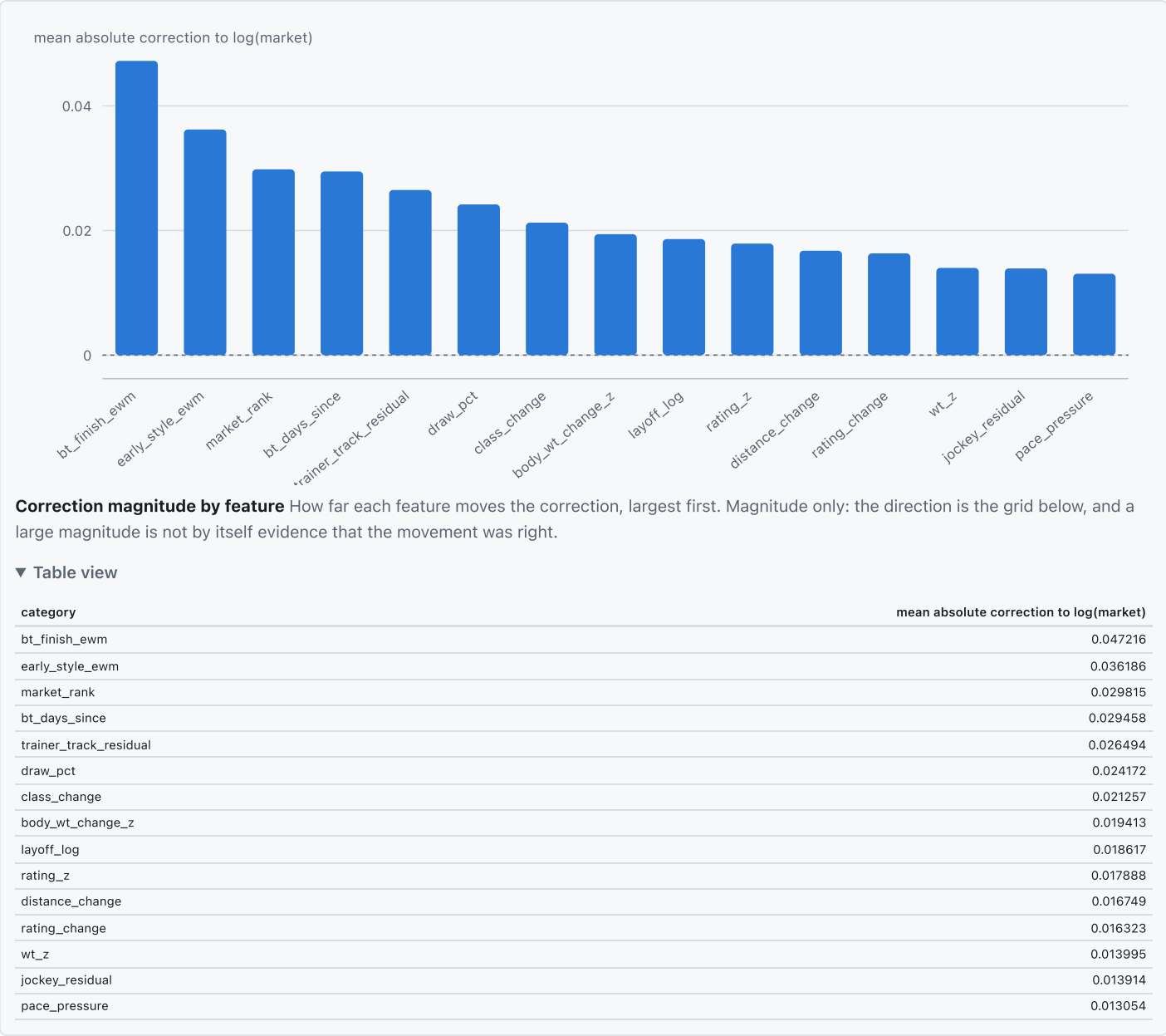
The booster learns a correction to `log(Implied_Prob)`, so a feature's attributed effect is its disagreement with the market -- not its contribution to a horse winning. Read the naive way, the sign inverts.

1 feature cannot be tested this way.

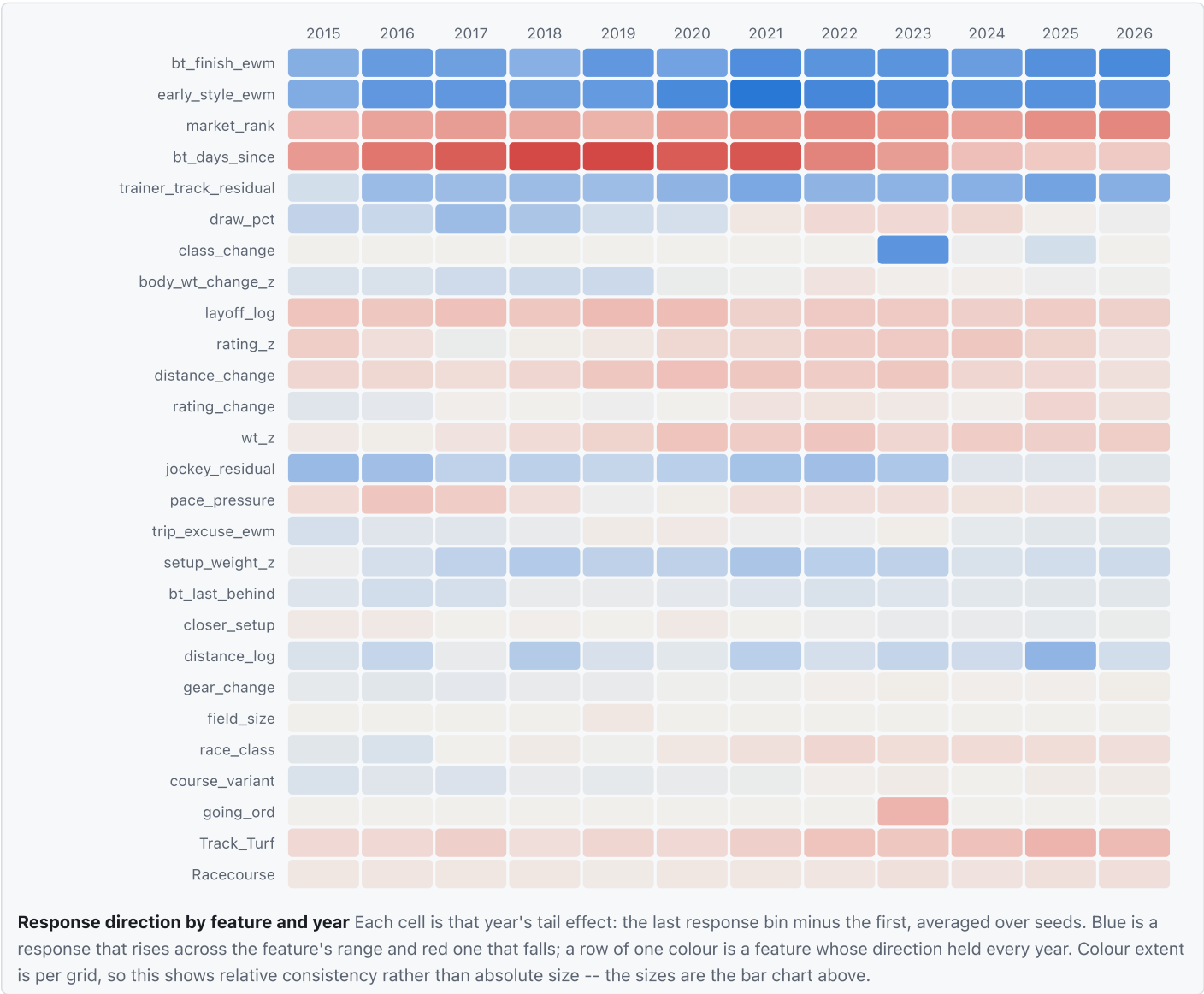
`market_rank` belongs to no group, so no ablation can remove it and this method cannot measure its held-out value.

label	feature	mean abs correction	sign agreement	rank stability	seed variation	psi	practical support	feature type	n years
stable_nonlinear	bt_finish_ewm	0.047216	0.991667	0.775991	0.058443	0.046310	12	continuous	12
stable_positive	early_style_ewm	0.036186	0.983333	0.570197	0.075225	0.007011	12	continuous	12
stable_nonlinear	market_rank	0.029815	1	0.461308	0.104674	0.001269	12	continuous	12
stable_nonlinear	bt_days_since	0.029458	0.833333	0.248826	0.100496	0.093612	12	continuous	12
stable_positive	trainer_track_residual	0.026494	0.950000	0.241158	0.080158	0.129261	12	continuous	12
stable_nonlinear	draw_pct	0.024172	0.925000	0.373624	0.120329	0.000533	12	continuous	12
stable_nonlinear	class_change	0.021257	0.766667	0.359226	0.266349	0.006694	12	categorical	12
stable_nonlinear	body_wt_change_z	0.019413	0.950000	0.238410	0.131227	0.010553	12	continuous	12
stable_nonlinear	layoff_log	0.018617	0.941667	0.322228	0.114803	0.103384	12	continuous	12
stable_nonlinear	rating_z	0.017888	0.925000	0.289509	0.249088	0.021924	12	continuous	12
stable_nonlinear	distance_change	0.016749	0.958333	0.193844	0.167273	0.019978	12	continuous	12
stable_nonlinear	rating_change	0.016323	0.950000	0.313155	0.193168	0.045097	12	continuous	12
stable_nonlinear	wt_z	0.013995	0.958333	0.272166	0.161691	0.035055	12	continuous	12
distribution_shifted	jockey_residual	0.013914	0.833333	0.227441	0.126905	0.794525	12	continuous	12
stable_nonlinear	pace_pressure	0.013054	0.885417	0.193571	0.144272	0.064975	12	continuous	12
stable_nonlinear	trip_excuse_ewm	0.012064	0.796296	0.268475	0.171276	0.058036	12	continuous	12

label	feature	mean abs correction	sign agreement	rank stability	seed variation	psi	practical support	feature type	n years
stable_nonlinear	setup_weight_z	0.011772	0.833333	0.446406	0.272290	0.002993	12	continuous	12
stable_nonlinear	bt_last_behind	0.010384	0.722222	0.200916	0.198664	0.026302	12	continuous	12
regime_dependent	closer_setup	0.005685	0.733333	0.775991	0.210115	0.057756	12	continuous	12
stable_nonlinear	distance_log	0.003923	1	0.616682	0.190325	0.072032	12	categorical	12
regime_dependent	gear_change	0.003599	0.583333	0.258334	0.175974	0.018293	6	binary	12
stable_nonlinear	field_size	0.002224	0.833333	0.479933	0.288189	0.106798	12	categorical	12
stable_nonlinear	race_class	0.002183	1	0.464102	0.334889	0.055002	12	categorical	12
stable_nonlinear	course_variant	0.001801	1	0.395644	0.125393	0.008740	12	categorical	12
distribution_shifted	going_ord	0.001708	0.591667	0.344006	0.451781	0.355145	12	categorical	12
stable_negative	Track_Turf	0.001099	1	0.500951	0.384960	0.004738	12	binary	12
stable_negative	Racecourse	0.001089	1	0.516146	0.108297	0.002257	12	binary	12



DIRECTION BY FEATURE AND YEAR



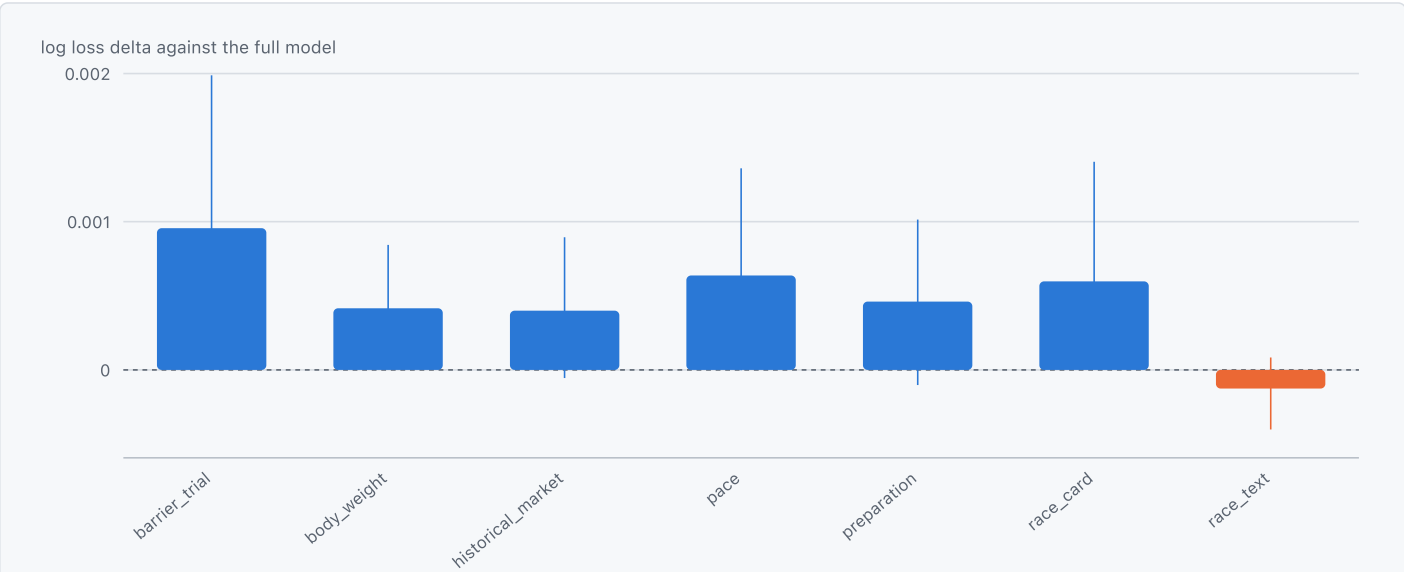
feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
bt_finish_ewm	0.004567	0.005879	0.005539	0.004393	0.006101	0.005372	0.006721	0.006365	0.006343	0.005667	0.006625	0.006990
early_style_ewm	0.004699	0.006102	0.006103	0.005494	0.005942	0.006982	0.008137	0.007150	0.006585	0.006344	0.006476	0.006303
market_rank	-0.002592	-0.003633	-0.003965	-0.003431	-0.002948	-0.003887	-0.004359	-0.004860	-0.004379	-0.003914	-0.004599	-0.004939
bt_days_since	-0.004123	-0.005793	-0.006845	-0.007643	-0.007718	-0.006902	-0.007143	-0.005097	-0.003971	-0.002321	-0.001852	-0.001765
trainer_track_residual	0.001226	0.003701	0.003620	0.003604	0.003463	0.004160	0.004985	0.004174	0.004264	0.004375	0.005257	0.004503
draw_pct	0.001928	0.001686	0.003571	0.002926	0.001287	0.001111	-0.000415	-0.001017	-0.001093	-0.001178	-0.000049	0.000157
class_change	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.006293	0.000161	0.001214	n/a
body_wt_change_z	0.000992	0.000999	0.001379	0.001441	0.001527	0.000218	0.000072	-0.000600	-0.000116	-0.000117	0.000190	0.000110
layoff_log	-0.002150	-0.001927	-0.002201	-0.001947	-0.002509	-0.002468	-0.001473	-0.001810	-0.001824	-0.001564	-0.001732	-0.001472
rating_z	-0.001628	-0.000783	0.000273	-0.000142	-0.000444	-0.001117	-0.001160	-0.001683	-0.001776	-0.001982	-0.001382	-0.000556
distance_change	-0.001252	-0.001107	-0.000867	-0.001204	-0.001946	-0.002259	-0.001937	-0.001747	-0.001920	-0.001218	-0.001026	-0.000703
rating_change	0.000765	0.000513	-0.000065	0.000030	0.000146	0.000029	-0.000600	-0.000621	-0.000353	-0.000106	-0.001267	-0.000752
wt_z	-0.000334	-0.000173	-0.000536	-0.000985	-0.001417	-0.002134	-0.001731	-0.002035	-0.001243	-0.001686	-0.001587	-0.001601
jockey_residual	0.003736	0.003379	0.002003	0.002173	0.001792	0.002275	0.003196	0.003416	0.002877	0.000741	0.000756	0.000680
pace_pressure	-0.000959	-0.002033	-0.001709	-0.000799	0.000126	-0.000129	-0.000853	-0.000843	-0.000807	-0.000666	-0.000548	-0.000721
trip_excuse_ewm	0.001115	0.000692	0.000702	0.000317	-0.000257	-0.000350	0.000164	0.000134	-0.000178	0.000384	0.000667	0.000666
setup_weight_z	0.000163	0.001139	0.002068	0.002580	0.002121	0.002113	0.002905	0.002280	0.002098	0.000942	0.001298	0.001441
bt_last_behind	0.000780	0.001340	0.001168	0.000315	0.000320	0.000471	0.000813	0.000989	0.000844	0.000603	0.000648	0.000627
closer_setup	-0.000352	-0.000326	0.000013	-0.000052	0.000031	-0.000358	0.000025	0.000179	0.000296	0.000337	0.000422	0.000270
distance_log	0.000949	0.001779	0.000342	0.002549	0.001079	0.000567	0.002247	0.001137	0.001852	0.001370	0.004053	0.001323
gear_change	0.000382	0.000650	0.000509	0.000349	0.000321	0.000108	0.000068	-0.000070	-0.000124	-0.000120	-0.000118	-0.000184
field_size	n/a	n/a	n/a	n/a	-0.000374	n/a	n/a	n/a	n/a	n/a	n/a	n/a

feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
race_class	0.000618	0.001006	0.000030	-0.000275	0.000077	-0.000443	-0.000724	-0.001189	-0.000933	-0.001000	-0.000867	-0.000778
course_variant	0.000972	0.000720	0.000902	0.000350	0.000418	0.000302	0.000262	-0.000075	-0.000245	-0.000036	-0.000250	-0.000242
going_ord	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	-0.002822	n/a	n/a	n/a
Track_Turf	-0.001057	-0.001087	-0.001573	-0.000829	-0.001225	-0.001056	-0.001510	-0.002079	-0.001971	-0.002186	-0.002848	-0.002545
Racecourse	-0.000438	-0.000496	-0.000439	-0.000383	-0.000242	-0.000384	-0.000419	-0.000492	-0.000762	-0.000546	-0.000703	-0.000812

This grid covers every year the run fitted, including any the evaluation population excludes. A learned response direction is a property of that year's fit rather than of the race population the headline metrics are computed on, and its per-feature mean is the stability table's `tail_effect` exactly -- so restricting one and not the other would have two parts of this report describe different populations. 15 of 27 features keep the same sign in every year they appear in. For the seven features that are constant within a race -- a racecourse, a surface -- the cell is the correction's magnitude and not its direction, because a race-constant feature cancels from every pairwise utility difference and has no direction to report. Which basis a row uses is in its tooltip and in the stability table's `curve_basis` column. A feature whose row changes colour is not necessarily broken: it may genuinely matter differently in different years. But its mean effect is then a summary of disagreeing years, and the mean is the least informative number about it.

GROUP ABLATION

group	mean delta vs full	bootstrap lower	bootstrap upper	n years	n seeds
barrier_trial	0.000956	0.000279	0.001988	12	5
body_weight	0.000415	0.000178	0.000844	12	5
historical_market	0.000400	-0.000055	0.000895	12	5
pace	0.000637	-0.000003	0.001360	12	5
preparation	0.000461	-0.000102	0.001015	12	5
race_card	0.000598	0.000150	0.001404	12	5
race_text	-0.000126	-0.000402	0.000083	12	5



Held-out value by feature group Each group removed and the model refitted, scored held out, with its bootstrap interval. Positive means removing the group made the model worse, so the group was carrying value. A bar whose interval spans zero was not distinguishable from noise.

▼ Table view

category	log loss delta against the full model	bootstrap interval
barrier_trial	0.000956	0.000279 to 0.001988
body_weight	0.000415	0.000178 to 0.000844
historical_market	0.0004	-0.000055 to 0.000895
pace	0.000637	-0.000003 to 0.00136
preparation	0.000461	-0.000102 to 0.001015
race_card	0.000598	0.00015 to 0.001404
race_text	-0.000126	-0.000402 to 0.000083

The deltas do *not* sum to the whole: features share information, so attributing a total by adding these up overstates it.

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

Role	What it is for
Primary	what an operator would run: sized for the rebate, and credited it.
Sizing reference	the same settlement, sized blind. It isolates sizing for the cliff from receiving the rebate -- this row and the primary settle identically and differ only in what the solver knew.
Settlement ablation	blind sizing with the rebate not credited. It isolates the rebate's own contribution to the return.
Benchmark	a fixed flat stake above an edge floor. Not a Kelly fraction, so it belongs to no preset panel below.

The table below is ordered by that role rather than alphabetically, so the first Kelly rows are the ones an operator would run.

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
rebate_aware_kelly@0.01	339	155,100	-10412.500000	-0.067134	-0.000742	0.004281	9992587.500000	325
rebate_aware_kelly@0.02	371	1,083,220	50671.500000	0.046779	0.011842	0.016956	10119124.500000	348
rebate_aware_kelly@0.05	772	7,045,400	-742,039	-0.105322	-0.017787	0.057885	9,823,706	728
rebate_aware_kelly@0.1	1398	17,945,530	-1,174,968	-0.065474	0.028644	0.085645	10,290,581	1302
kelly_settled_rebate@0.01	338	110,870	-578	-0.005213	-0.000058	0.002609	9,999,422	324
kelly_settled_rebate@0.02	341	221,750	-947.500000	-0.004273	-0.000095	0.005229	9999052.500000	327
kelly_settled_rebate@0.05	346	554,210	-2545	-0.004592	-0.000255	0.013015	9,997,455	332
kelly_settled_rebate@0.1	346	1,109,420	-6444.500000	-0.005809	0.001113	0.025355	10011131.500000	332
kelly_no_rebate@0.01	338	110,870	-578	-0.005213	-0.000058	0.002609	9,999,422	324
kelly_no_rebate@0.02	341	221,750	-947.500000	-0.004273	-0.000095	0.005229	9999052.500000	327
kelly_no_rebate@0.05	346	554,210	-2545	-0.004592	-0.000255	0.013015	9,997,455	332
kelly_no_rebate@0.1	346	1,108,400	-6718	-0.006061	-0.000672	0.025854	9,993,282	332
flat_stake@edge>=0.02	174	1,740,000	-973,500	-0.559483	-0.102420	0.098882	9,026,500	169

Model view bagged_offset. Seed -1. Bankroll scenario hkd_10m, opening HK\$10,000,000 (primary). These rules see the same races and the same odds. They do not necessarily choose the same runners or the same races: rebate-aware sizing admits runners a blind solve would reject, because staking a mildly negative-edge runner raises the loss on every other outcome and can push a race over the rebate threshold. Compare the bet counts as well as the returns.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

Bankroll scenario hkd_10m, opening HK\$10,000,000 (primary).

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.031990	0.031990	-0.6870	115
0.050000 to 0.100000	0.069152	0.069152	-0.5160	50
0.100000 to 0.200000	0.133063	0.133063	0.8278	9

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009823	0.009823	0.0105	0.0161	0.0159	0.0155	164	167	172	172
0.020000 to 0.050000	0.031990	0.031990	-0.4455	-0.4476	-0.4485	-0.4513	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	-0.0831	-0.0821	-0.0834	-0.0863	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	3.6013	3.6146	3.6301	9	9	9	9

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009823	0.009823	0.0105	0.0161	0.0159	0.0155	164	167	172	172
0.020000 to 0.050000	0.031990	0.031990	-0.4455	-0.4476	-0.4485	-0.4512	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	-0.0831	-0.0821	-0.0834	-0.0859	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	3.6013	3.6146	3.6321	9	9	9	9

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.005611	-0.005611	0.8000	0.1869	-0.0671	-0.0220	1	30	427	1053
0 to 0.020000	0.009823	0.009823	-0.5201	-0.2768	-0.1540	-0.1224	164	167	171	171
0.020000 to 0.050000	0.031990	0.031990	-0.4459	-0.2440	-0.4412	-0.4780	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	0.1596	0.1275	-0.1227	-0.1348	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	5.4892	4.0988	3.6541	9	9	9	9

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL

HK\$10,000,000

scenario hkd_10m

PRESETS SETTLED

1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly

PATH MODES

annual reset, continuous

SEEDS

5

7, 17, 42, 123, 256

TRAINED ON

initial 2010-2015, expanding by one season per evaluation year

a separate fit per evaluation year, each on 2010 through the year before it -- 11 fits in all

BOOSTING ROUNDS

400

YEARS STAKED

2016-2026

2015 excluded

BANKROLL SCENARIOS

2

hkd_10m, hkd_1b; each solved independently

REBATE THRESHOLD

1e-03 of the bank

a fixed HK\$10,000, so its bankroll fraction is what makes two scenarios differ

The 2016 evaluation model was trained on the initial 2010-2015 set. Each later evaluation adds the preceding complete season, so the training window expands by one year; the 2026 model uses 2010-2025. Every evaluation year is fitted separately -- 11 fits in all -- and no evaluation year's outcomes appear in its own fit. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. annual-reset re-settlement of continuous-path decisions: the same stakes, compounded from the declared opening balance at the start of each test year. Not an annual-reset sizing simulation -- stakes are not re-solved from reset capital, and doing so would change which bets are placed rather than only how they compound. Each bankroll scenario below is solved from its own opening balance, sequentially by race, because the rebate threshold is a fixed dollar amount and the fraction the solver sees therefore changes with the bank. The larger scenario is not the smaller one's stakes scaled up. 2015 is excluded from every path below. That year's utility scale fell back to its theoretical value rather than being fitted, so its stakes are not comparable with the rest.

HK\$10.0M, 1% KELLY

OPENING BANKROLL

HK\$10,000,220

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

-0.076%

Rebate-aware Kelly, continuous

BETS

336

settled bets

REBATE RECEIVED

HK\$3,000

by the primary rule

Rebate-aware Kelly Kelly, settled rebate (= Kelly, no rebate)



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	329	+0.00284%	336 bets
Rebate-aware Kelly, continuous	329	-0.076%	336 bets
Kelly, settled rebate, annual reset	328	+0.00284%	335 bets
Kelly, settled rebate, continuous	328	-0.00798%	335 bets
Kelly, no rebate, annual reset	328	+0.00284%	335 bets
Kelly, no rebate, continuous	328	-0.00798%	335 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	14	14	14
2017	37	37	37
2018	42	42	42
2019	49	49	49
2020	59	58	58
2021	54	54	54
2022	37	37	37
2023	21	21	21
2024	10	10	10
2025	11	11	11
2026	2	2	2

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 14 bets in 2016 against 11 in 2025. Kelly, settled rebate and Kelly, no rebate: 14 bets in 2016 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, settled rebate and Kelly, no rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, settled rebate and Kelly, no rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK\$3,000; Rebate-aware Kelly, continuous HK\$3,000. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$10.0M, 2% KELLY

OPENING BANKROLL

HK\$10,000,420

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

+1.2%

Rebate-aware Kelly, continuous

BETS

368

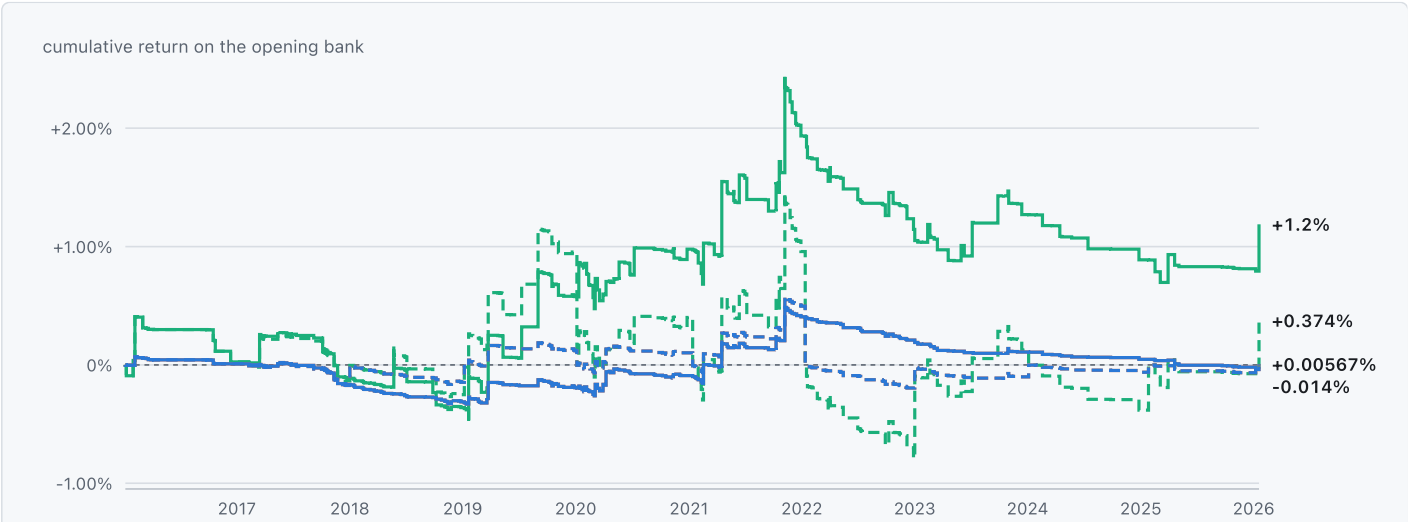
settled bets

REBATE RECEIVED

HK\$68,453

by the primary rule

Rebate-aware Kelly Kelly, settled rebate (= Kelly, no rebate)



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	360	+0.374%	368 bets
Rebate-aware Kelly, continuous	360	+1.2%	368 bets
Kelly, settled rebate, annual reset	331	+0.00567%	338 bets

BETS PER YEAR

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 17 bets in 2016 against 13 in 2025. Kelly, settled rebate and Kelly, no rebate: 15 bets in 2016 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, settled rebate and Kelly, no rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, settled rebate and Kelly, no rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK\$68,453; Rebate-aware Kelly, continuous HK\$68,453. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$10.0M, 5% KELLY

HK\$9,994,880

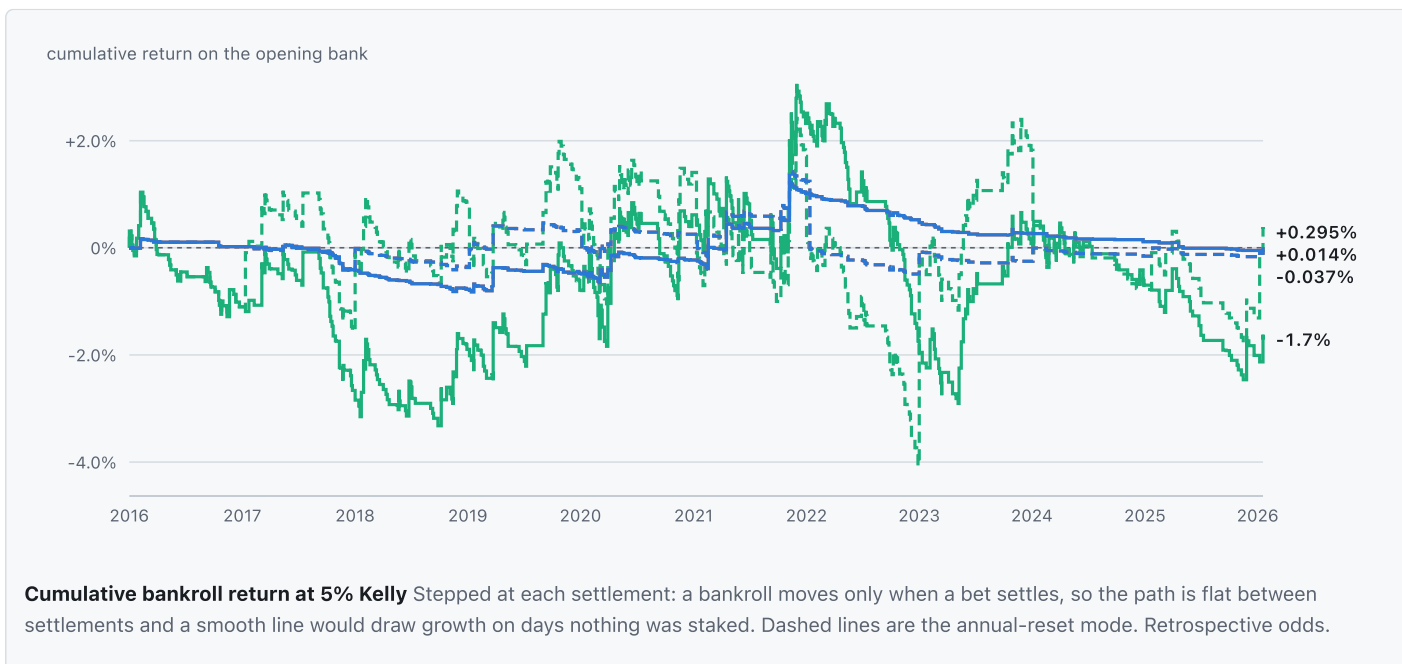
-1.7%

756

HK\$552,745

by the primary rule

■ Rebate-aware Kelly ■ Kelly, settled rebate (= Kelly, no rebate)



▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	724	+0.295%	756 bets
Rebate-aware Kelly, continuous	724	-1.7%	756 bets
Kelly, settled rebate, annual reset	336	+0.014%	343 bets
Kelly, settled rebate, continuous	336	-0.037%	343 bets
Kelly, no rebate, annual reset	336	+0.014%	343 bets
Kelly, no rebate, continuous	336	-0.037%	343 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	50	16	16
2017	72	39	39
2018	80	42	42
2019	77	50	50
2020	110	58	58
2021	124	54	54
2022	98	39	39
2023	61	22	22
2024	44	10	10
2025	37	11	11
2026	3	2	2

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 50 bets in 2016 against 37 in 2025. Kelly, settled rebate and Kelly, no rebate: 16 bets in 2016 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, settled rebate and Kelly, no rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, settled rebate and Kelly, no rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK\$552,745; Rebate-aware Kelly, continuous HK\$552,745. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$10.2M, 10% KELLY

OPENING BANKROLL

HK\$10,179,936

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

+1.1%

Rebate-aware Kelly, continuous

BETS

1,354

settled bets

REBATE RECEIVED

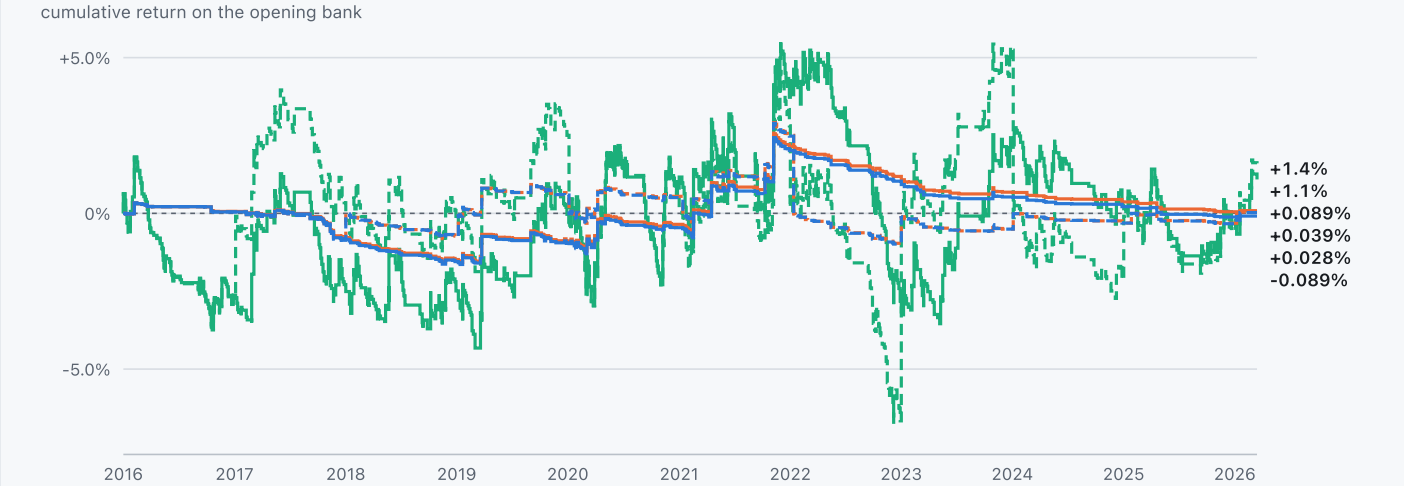
HK\$1,428,473

by the primary rule

Rebate-aware Kelly

Kelly, settled rebate

Kelly, no rebate



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	1,280	+1.4%	1,354 bets
Rebate-aware Kelly, continuous	1,280	+1.1%	1,354 bets
Kelly, settled rebate, annual reset	336	+0.039%	343 bets
Kelly, settled rebate, continuous	336	+0.089%	343 bets
Kelly, no rebate, annual reset	336	+0.028%	343 bets
Kelly, no rebate, continuous	336	-0.089%	343 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	105	16	16
2017	121	39	39
2018	138	42	42
2019	117	50	50
2020	188	58	58
2021	198	54	54
2022	186	39	39
2023	112	22	22
2024	87	10	10
2025	86	11	11
2026	16	2	2

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 105 bets in 2016 against 86 in 2025. Kelly, settled rebate and Kelly, no rebate: 16 bets in 2016 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$1,428,473; Rebate-aware Kelly, continuous HK\$1,428,473; Kelly, settled rebate, annual reset HK\$17,576; Kelly, settled rebate, continuous HK\$17,576. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

BANKROLL-SCALE SENSITIVITY

Opening bank	Role	Fraction	Final return	Bets	Rebate received
HK\$10.0M	Settlement ablation	1% Kelly	-0.00798%	335	HK\$0
HK\$10.0M	Sizing reference	1% Kelly	-0.00798%	335	HK\$0
HK\$10.0M	Primary	1% Kelly	-0.076%	336	HK\$3,000
HK\$10.0M	Settlement ablation	2% Kelly	-0.014%	338	HK\$0
HK\$10.0M	Sizing reference	2% Kelly	-0.014%	338	HK\$0

Opening bank	Role	Fraction	Final return	Bets	Rebate received
HK\$10.0M	Primary	2% Kelly	+1.2%	368	HK\$68,453
HK\$10.0M	Settlement ablation	5% Kelly	-0.037%	343	HK\$0
HK\$10.0M	Sizing reference	5% Kelly	-0.037%	343	HK\$0
HK\$10.0M	Primary	5% Kelly	-1.7%	756	HK\$552,745
HK\$10.0M	Settlement ablation	10% Kelly	-0.089%	343	HK\$0
HK\$10.0M	Sizing reference	10% Kelly	+0.089%	343	HK\$17,576
HK\$10.2M	Primary	10% Kelly	+1.1%	1,354	HK\$1,428,473
HK\$1.0B	Settlement ablation	1% Kelly	-0.00608%	343	HK\$0
HK\$1.0B	Sizing reference	1% Kelly	+0.079%	343	HK\$848,141
HK\$1.0B	Primary	1% Kelly	+0.846%	3,040	HK\$16,086,880
HK\$1.0B	Settlement ablation	2% Kelly	-0.013%	343	HK\$0
HK\$1.0B	Sizing reference	2% Kelly	+0.164%	343	HK\$1,766,300
HK\$1.0B	Primary	2% Kelly	+1.5%	3,245	HK\$32,472,529
HK\$1.0B	Settlement ablation	5% Kelly	-0.037%	343	HK\$0
HK\$1.0B	Sizing reference	5% Kelly	+0.410%	343	HK\$4,466,977
HK\$1.0B	Primary	5% Kelly	+3.7%	3,413	HK\$83,383,236
HK\$1.0B	Settlement ablation	10% Kelly	-0.089%	343	HK\$0
HK\$1.0B	Sizing reference	10% Kelly	+0.808%	343	HK\$8,976,235
HK\$1.0B	Primary	10% Kelly	+7.1%	3,467	HK\$174,159,010

Each scenario is solved independently and sequentially by race, because the rebate threshold is a fixed HK\$10,000 and the fraction the solver sees is therefore 1e-3 of a HK\$10M bank and 1e-5 of a HK\$1B one. These are two different optimization problems; the larger bank's stakes are not the smaller bank's scaled up.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,000,284	+0.00284%	335	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$10,000,220	HK\$9,999,422	-0.00798%	335	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,000,284	+0.00284%	335	HK\$0
1% Kelly	Kelly, settled rebate, continuous	HK\$10,000,220	HK\$9,999,422	-0.00798%	335	HK\$0
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,000,284	+0.00284%	336	HK\$3,000
1% Kelly	Rebate-aware Kelly, continuous	HK\$10,000,220	HK\$9,992,588	-0.076%	336	HK\$3,000
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,000,567	+0.00567%	338	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$10,000,420	HK\$9,999,052	-0.014%	338	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,000,567	+0.00567%	338	HK\$0
2% Kelly	Kelly, settled rebate, continuous	HK\$10,000,420	HK\$9,999,052	-0.014%	338	HK\$0
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,037,440	+0.374%	368	HK\$68,453
2% Kelly	Rebate-aware Kelly, continuous	HK\$10,000,420	HK\$10,119,124	+1.2%	368	HK\$68,453
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,001,408	+0.014%	343	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$10,001,126	HK\$9,997,455	-0.037%	343	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,001,408	+0.014%	343	HK\$0
5% Kelly	Kelly, settled rebate, continuous	HK\$10,001,126	HK\$9,997,455	-0.037%	343	HK\$0
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,029,518	+0.295%	756	HK\$552,745
5% Kelly	Rebate-aware Kelly, continuous	HK\$9,994,880	HK\$9,823,706	-1.7%	756	HK\$552,745
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,002,815	+0.028%	343	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$10,002,186	HK\$9,993,282	-0.089%	343	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,003,868	+0.039%	343	HK\$17,576
10% Kelly	Kelly, settled rebate, continuous	HK\$10,002,186	HK\$10,011,132	+0.089%	343	HK\$17,576
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,144,542	+1.4%	1,354	HK\$1,428,473
10% Kelly	Rebate-aware Kelly, continuous	HK\$10,179,936	HK\$10,290,581	+1.1%	1,354	HK\$1,428,473

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

DAY-BLOCK BOOTSTRAP OF THE BANKROLL

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02 / block1	2000	HK\$9,011,645	HK\$8,701,055	1.0000	0.1001	0.1304	0.0000
flat_stake@edge>=0.02 / block3	2000	HK\$9,027,154	HK\$8,722,326	1.0000	0.0988	0.1292	0.0000
flat_stake@edge>=0.02 / block7	2000	HK\$9,022,940	HK\$8,676,218	0.9995	0.0992	0.1339	0.0000
kelly_no_rebate@0.01 / block1	2000	HK\$9,998,828	HK\$9,962,064	0.5175	0.0027	0.0042	0.0000
kelly_no_rebate@0.01 / block3	2000	HK\$9,997,755	HK\$9,965,458	0.5385	0.0027	0.0039	0.0000
kelly_no_rebate@0.01 / block7	2000	HK\$9,998,701	HK\$9,968,259	0.5265	0.0025	0.0037	0.0000
kelly_no_rebate@0.02 / block1	2000	HK\$9,997,322	HK\$9,925,704	0.5230	0.0054	0.0083	0.0000
kelly_no_rebate@0.02 / block3	2000	HK\$9,997,433	HK\$9,937,364	0.5220	0.0052	0.0073	0.0000
kelly_no_rebate@0.02 / block7	2000	HK\$9,999,750	HK\$9,948,096	0.5035	0.0051	0.0067	0.0000
kelly_no_rebate@0.05 / block1	2000	HK\$9,992,713	HK\$9,818,800	0.5275	0.0135	0.0205	0.0000
kelly_no_rebate@0.05 / block3	2000	HK\$9,997,740	HK\$9,838,260	0.5055	0.0131	0.0187	0.0000
kelly_no_rebate@0.05 / block7	2000	HK\$9,996,284	HK\$9,828,915	0.5145	0.0125	0.0201	0.0000
kelly_no_rebate@0.1 / block1	2000	HK\$9,983,809	HK\$9,639,911	0.5290	0.0269	0.0406	0.0000
kelly_no_rebate@0.1 / block3	2000	HK\$9,993,867	HK\$9,678,171	0.5075	0.0260	0.0371	0.0000
kelly_no_rebate@0.1 / block7	2000	HK\$9,990,857	HK\$9,659,398	0.5175	0.0249	0.0399	0.0000
kelly_settled_rebate@0.01 / block1	2000	HK\$9,998,828	HK\$9,962,064	0.5175	0.0027	0.0042	0.0000
kelly_settled_rebate@0.01 / block3	2000	HK\$9,997,755	HK\$9,965,458	0.5385	0.0027	0.0039	0.0000
kelly_settled_rebate@0.01 / block7	2000	HK\$9,998,701	HK\$9,968,259	0.5265	0.0025	0.0037	0.0000
kelly_settled_rebate@0.02 / block1	2000	HK\$9,997,322	HK\$9,925,704	0.5230	0.0054	0.0083	0.0000
kelly_settled_rebate@0.02 / block3	2000	HK\$9,997,433	HK\$9,937,364	0.5220	0.0052	0.0073	0.0000
kelly_settled_rebate@0.02 / block7	2000	HK\$9,999,750	HK\$9,948,096	0.5035	0.0051	0.0067	0.0000
kelly_settled_rebate@0.05 / block1	2000	HK\$9,992,713	HK\$9,818,800	0.5275	0.0135	0.0205	0.0000
kelly_settled_rebate@0.05 / block3	2000	HK\$9,997,740	HK\$9,838,260	0.5055	0.0131	0.0187	0.0000
kelly_settled_rebate@0.05 / block7	2000	HK\$9,996,284	HK\$9,828,915	0.5145	0.0125	0.0201	0.0000
kelly_settled_rebate@0.1 / block1	2000	HK\$10,001,580	HK\$9,656,484	0.4975	0.0262	0.0388	0.0000
kelly_settled_rebate@0.1 / block3	2000	HK\$10,012,580	HK\$9,697,626	0.4775	0.0253	0.0356	0.0000
kelly_settled_rebate@0.1 / block7	2000	HK\$10,007,943	HK\$9,677,738	0.4845	0.0242	0.0384	0.0000
rebate_aware_kelly@0.01 / block1	2000	HK\$9,992,109	HK\$9,940,904	0.5870	0.0042	0.0068	0.0000
rebate_aware_kelly@0.01 / block3	2000	HK\$9,993,762	HK\$9,947,417	0.5905	0.0040	0.0065	0.0000
rebate_aware_kelly@0.01 / block7	2000	HK\$9,992,879	HK\$9,952,480	0.6110	0.0041	0.0061	0.0000
rebate_aware_kelly@0.02 / block1	2000	HK\$10,126,283	HK\$9,846,592	0.2430	0.0171	0.0282	0.0000
rebate_aware_kelly@0.02 / block3	2000	HK\$10,123,226	HK\$9,869,781	0.2250	0.0164	0.0263	0.0000
rebate_aware_kelly@0.02 / block7	2000	HK\$10,114,399	HK\$9,867,028	0.2225	0.0147	0.0250	0.0000
rebate_aware_kelly@0.05 / block1	2000	HK\$9,816,230	HK\$8,973,151	0.6355	0.0728	0.1206	0.0000
rebate_aware_kelly@0.05 / block3	2000	HK\$9,806,013	HK\$8,937,516	0.6405	0.0752	0.1240	0.0000
rebate_aware_kelly@0.05 / block7	2000	HK\$9,843,809	HK\$8,985,885	0.6000	0.0710	0.1187	0.0000
rebate_aware_kelly@0.1 / block1	2000	HK\$10,255,008	HK\$8,553,161	0.4205	0.1223	0.2099	0.0000
rebate_aware_kelly@0.1 / block3	2000	HK\$10,241,534	HK\$8,504,669	0.4165	0.1239	0.2176	0.0000
rebate_aware_kelly@0.1 / block7	2000	HK\$10,298,927	HK\$8,621,977	0.3920	0.1203	0.2038	0.0000

Race days resampled with replacement, so results from one meeting stay together. This is sampling risk in the settled path, not model risk. 39 of 234 settled series shown, filtered to model view bagged_offset.

MODEL-ASSUMED MONTE CARLO

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02	1000	HK\$10,001,943	HK\$9,203,076	0.4980	0.0514	0.0920	0.0000

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
kelly_no_rebate@0.01	1000	HK\$10,003,246	HK\$9,969,687	0.4390	0.0021	0.0041	0.0000
kelly_no_rebate@0.02	1000	HK\$10,007,155	HK\$9,938,587	0.4400	0.0043	0.0083	0.0000
kelly_no_rebate@0.05	1000	HK\$10,013,567	HK\$9,835,978	0.4590	0.0109	0.0206	0.0000
kelly_no_rebate@0.1	1000	HK\$10,025,462	HK\$9,673,977	0.4600	0.0217	0.0409	0.0000
kelly_settled_rebate@0.01	1000	HK\$10,003,246	HK\$9,969,687	0.4390	0.0021	0.0041	0.0000
kelly_settled_rebate@0.02	1000	HK\$10,007,155	HK\$9,938,587	0.4400	0.0043	0.0083	0.0000
kelly_settled_rebate@0.05	1000	HK\$10,013,567	HK\$9,835,978	0.4590	0.0109	0.0206	0.0000
kelly_settled_rebate@0.1	1000	HK\$10,025,428	HK\$9,673,827	0.4600	0.0217	0.0409	0.0000

Winners drawn from the model's own probabilities. 9 of 54 settled series shown, filtered to model view bagged_offset.

One simulation was declined rather than approximated

Refusals are recorded rather than omitted, so a reader can see the strategy was considered and declined. Rebate-aware sizing cannot reuse stale bankroll-dependent fractions on a divergent path.

The realised path, the day-block bootstrap and the model-assumed MC answer different questions and must not be shown as one uncertainty band.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix and provenance

CONFIGURATION

Item	Value
Market offset column	Implied_Prob
Features	28
Features the booster learns on	27
Opening bankroll	10,000,000
Bankroll scenario	hkd_10m
Scenario role	primary
Rebate threshold as a bankroll fraction	0.001000
Dataset registry digest	sha256:f93edc409844f3df108c777ef6f48931b6b0d095d5b9b16fb1ae1a590da08844

definition, raw_source, units, default_behavior, higher_value_meaning and known_caveats are intentionally blank. They are claims about the domain, not properties derivable from the model, and inventing them would produce documentation that reads as authoritative and is not.

KNOWN SOURCE-DATA DEFECTS

The lineage record carries {'landing_dataset': 'aspx-results', 'landing_duplicate_keys': 1513, 'landing_primary_key': ['race_id', 'Horse No.'], 'retained_key': ['race_id', 'Horse No.'], 'retained_policy': 'keep_first', 'retained_policy_source': 'features/build.py', 'feature_table_duplicate_keys': 0, 'absorbed_by_build': 1513, 'consistent': True}. Recorded here rather than corrected silently: the feature build absorbs these, and a reader auditing a number should know they exist.

PROVENANCE

Item	Value
Source commit	0422b20c2039
Source tree	dirty at generation
Config digest	sha256:c07c6a2a1a11d3454f31c509879a364fc5d9abb34347403758907f9bde0ddd10
Identity source	artifact
Model view	bagged_offset
Artifact root	results/model_report
Inputs read	19 artifact(s), locked by digest in run_lock.json

Item	Value
Evaluation race keys	7933 races, sha256:342d95ee674808a0d7ae4d0df5b0f958da1898863e25c2898bb71d9f58392e58
Environment: numpy	2.5.2
Environment: pandas	3.0.5
Environment: python	3.12.11
Environment: xgboost	3.4.1

The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

Every input this report read is listed in `run_lock.json` by content digest, together with the lock digest of the run that produced them -- which is what chains a published figure back through the report to the fit and from there to the feature table's own digest.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Softmax Offset against the market on the same races. It does not measure this model against the other published models except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.